CHENG-HSUAN LEE

Taipei, Taiwan | +886 905 499 682 | lee940750@gmail.com | lee940750.github.io

Robotics-focused Mechanical Engineering undergraduate | Legged robotics, robot learning, actuator design, sim-to-real testing

PROFILE AND RESEARCH DIRECTION

Mechanical Engineering undergraduate at National Taiwan University focused on building complete physical robot systems. My current work in NTUME ASR Lab connects 12-DoF quadruped locomotion, actuator/reducer design, motor communication, Isaac Lab reinforcement-learning workflows, and sim-to-real debugging. I am especially interested in legged robots whose hardware, control policies, and experimental behavior are designed together rather than treated as separate problems.

EDUCATION

National Taiwan University - B.S. in Mechanical Engineering

Taipei, Taiwan | Expected 2027

Relevant coursework: Automatic Control; Microprocessor Controlled Systems; Applied Electronics; Mechanism; Machine Design Theory; Engineering Graphics; Practice of Mechanical Engineering; Manufacturing Processes; Workshop Practice; Mechanical Engineering Laboratory I/II; Engineering Materials; Strength of Materials; Dynamics; Numerical Analysis; Fluid Mechanics; Thermodynamics; Heat Transfer.

Taipei Municipal Chien Kuo High School

Taipei, Taiwan | 2020-2023

Built early robotics foundation through CKHS Robotics Team, FRC programming, strategy, scouting, and international competition work.

RESEARCH AND LABORATORY EXPERIENCE

NTUME ASR Lab - Quadruped Robot Learning and Hardware Integration

Current research work | Legged robotics, robot learning, actuator design, hardware-control integration

Current focus: building the missing interfaces between a simulated quadruped policy and a physical robot. My work is not only to train a walking policy, but to make the hardware, actuator limits, communication loop, model assumptions, and safety checks consistent enough for real deployment.

- Developing a 12-DoF quadruped workflow that links mechanical architecture, leg module packaging, actuator layout, compact reducer concepts, wiring, embedded motor communication, simulation setup, and physical robot validation.
- Preparing robot models and deployment assumptions around URDF/model checks, joint limits, action scaling, observation mapping, command constraints, actuator saturation, and hardware-feasible motion.
- Working with NVIDIA Isaac Lab / Isaac Sim locomotion workflows, including DWAQ/PPO training, reward design, observation design, action-space tuning, gait stability, foot clearance, contact behavior, and sim-to-real mismatch analysis.
- Analyzing reward behavior and training logs when terms such as action-rate, torque, and joint-acceleration penalties become too weak, too strong, or numerically abnormal; using those signals to reason about gait smoothness and hardware stress.
- Iterating compact actuation concepts such as cycloidal and planetary reducer layouts while considering torque transmission, backlash, bearing support, manufacturability, assembly tolerance, wiring access, repairability, and repeated testing.
- Built motor-control diagnostics for speed and position commands with speed, position, and current feedback monitoring; used serial readback, timeout checks, command scaling, and calibration to separate software parsing issues from wiring or transceiver problems.
- Debugging hardware communication paths at the board and wiring level, including RS485 receive behavior, RX/A/B/GND checks, direction-control assumptions, and clean serial-monitor output for repeatable actuator tests.
- Practicing deployment-oriented debugging across ROS/Pinocchio-related workflows, policy export, actuator limits, safety checks, reset assumptions, mechanical compliance, timing/communication issues, and repeated on-robot validation.

Research methods	Isaac Lab / Isaac Sim, PyTorch-style RL workflows, DWAQ/PPO, reward shaping, observation/action-space tuning, URDF checks, sim-to-real debugging.
Hardware interface	Actuator/reducer packaging, motor speed/position/current feedback, Teensy/ESP32, RS485/UART/I2C/PWM, calibration, wiring-level failure isolation.

SELECTED ENGINEERING PROJECTS

Quadruped Robot Dog - Team Lead, Mechanical Designer, Controls Developer

Microprocessor Controlled Systems | A+ outcome | ESP32, PCA9685, MATLAB, Arduino, 3D-printed structure

- Designed and integrated a compact eight-servo quadruped platform around serviceability, wiring, battery placement, center-of-gravity position, and repeated repair during gait testing.
- Translated inverse-kinematics and gait reasoning from MATLAB into embedded ESP32/PCA9685 servo control, separating behavior logic, gait timing, calibration, and servo output for easier debugging.
- Demonstrated walking, standing, sitting, waving, hip-lift, and rocking behaviors, turning a course project into a visible legged-robotics foundation for later research work.

Quadcopter Ball-Transfer Drone - Team Lead and Systems Integrator

Practice of Mechanical Engineering | Full marks and A+ outcome | Peer reference point for many teams

- Led full-system mission design for a drone that had to fly, carry, transfer, and release balls reliably under final evaluation pressure.
- Coordinated airframe, payload mechanism, flight controller, SBUS receiver, ESC/motor setup, spare-system planning, and final-risk control; our architecture and testing process became a reference point for many peer teams.
- Used thrust-to-throttle tests, drift diagnosis, vibration checks, crash recovery, payload trials, and signal-chain debugging to convert failures into design decisions.

Foldable Ball Transfer Mechanism - Team Lead

Machine Design Theory | Mechanism design, packaging constraint, fabrication-aware decision making

- Designed a foldable long-distance ball-transfer mechanism under packaging and deployment constraints, balancing ideal motion with buildable structure.
- Evaluated folding architecture, linkage geometry, component accessibility, fabrication feasibility, and tolerance sensitivity before committing to the final design.

COMPETITION ROBOTICS AND LEADERSHIP

CKHS Robotics Team - Programming and Strategy Lead

FIRST Robotics Competition and scouting/strategy systems

- Led programming and strategy work in competition robotics, connecting autonomous logic, driver needs, mechanism limits, match strategy, and repair speed under real event pressure.
- Built scouting and analysis workflows, including mobile data collection and Excel/VBA-based comparison tools, to turn large match datasets into alliance-selection and strategy decisions.

AWARDS AND RECOGNITION

- FRC Sacramento Regional - Finalist and Industrial Design Award.
- FRC New Taipei City x Hon Hai Regional - Engineering Inspiration Award; qualified for Houston Championship.
- 2022 FRC Carver Championship Division, Houston - international championship participation.
- 2023 FRC Los Angeles Regional - Excellence in Engineering Award.
- 2023 FRC San Diego Regional - Excellence in Engineering Award.
- 2nd ISS Kibo Robot Programming Challenge Taiwan Preliminary - Merit Award.
- 58th Hsinchu County Science Fair, Biology Division - Second Place.

TECHNICAL SKILLS

Mechanical Systems	Mechanism design; actuator packaging; cycloidal/planetary reducer concepts; CAD; 3D printing; machining; engineering drawings; tolerance and assembly planning.
Robot Learning and Control	NVIDIA Isaac Lab / Isaac Sim; DWAQ/PPO workflows; reward shaping; observation/action-space tuning; URDF; ROS; Pinocchio; MATLAB/Simulink; motor calibration.
Embedded and Software	Python; C++; Arduino; ESP32; Teensy 4.0; PCA9685; RS485; UART; I2C; PWM; Git/GitHub; Linux; VS Code; serial diagnostics.
Testing and Documentation	Thrust testing; vibration and drift diagnosis; motor speed/position/current feedback; waveform checks; failure isolation; engineering reports; portfolio documentation.

TECHNICAL DOCUMENTATION AND PORTFOLIO

Portfolio website: lee940750.github.io. Includes robotics case studies, project videos/photos, translated technical reports, and supporting coursework evidence. Application materials are organized for fast faculty review, with CV and major project reports available from the Reports PDF section.